

Team 4A

The Breathing Light

CPE190/EEE193A

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I. Abstract

Stress is a problem that many people will experience at some point in their lives. Due to various circumstances, it can be difficult to manage this stress, usually requiring long complicated procedures to limit and relieve it. Due to poor stress management, mental and physical performance drastically decrease, and an assortment of mental and physical health issues can occur. Of course, everyone has their go-to reliever, but there still seems to be an overabundance of stress in our culture. However, we believe a crucial way to relieve stress is through the regular use of deep, diaphragmatic breathing that can easily be learned and used by everyone. It is our goal, with the knowledge collected on the issue of stress, to develop a light-based system with sound and haptic feedback to reduce stress and improve mental well-being by subtly guiding the users breathing.

II. Introduction

A. Societal Problem

Stress plays a role in all demographics of people across the globe. It can easily be left unaddressed during daily life, but too much stress can cause all sorts of mental and physical problems to occur. A way to limit the effects of stress is by breathing techniques which are applicable to everyone regardless of demographic. According to the ADA, anxiety disorders such as social anxiety, OCD, and PTSD affect millions of people globally. The buildup of stress affects different aspects of our daily lives such as mood, work performance, focus, and sleep. Habits such as excessive use of electronics before bed, answering calls/texts throughout the night, and poor diet are responsible for buildup of stress. Sleep plays an important role in recovering, repairing, and energizing our bodies. With poor stress management, sleeping becomes impaired which leads to several collapses in our

daily lives. Psychological stress can even have a physical toll on our bodies. It has been reported that patients who suffer from psychological stress can have symptoms associated with infections and tissue damage.

The rampancy of chronic stress raises many ethical concerns, and it is our belief that this issue should not be left unsolved. Using the evidence and studies researched, it is our goal to design a device to help people relieve stress. This device should be something that anyone can use and benefit greatly from, which is why breathing techniques were chosen. Breathing is natural for everyone and is easily guided, therefore little to no cognition is needed to perform stress reducing, diaphragmatic breathing when an outside source is guiding it. Our system aims to subtly guide the users into slowing their breathing, which we believe will slow their heart rate, increase oxygen levels in the blood, and activate the body's natural healing process.

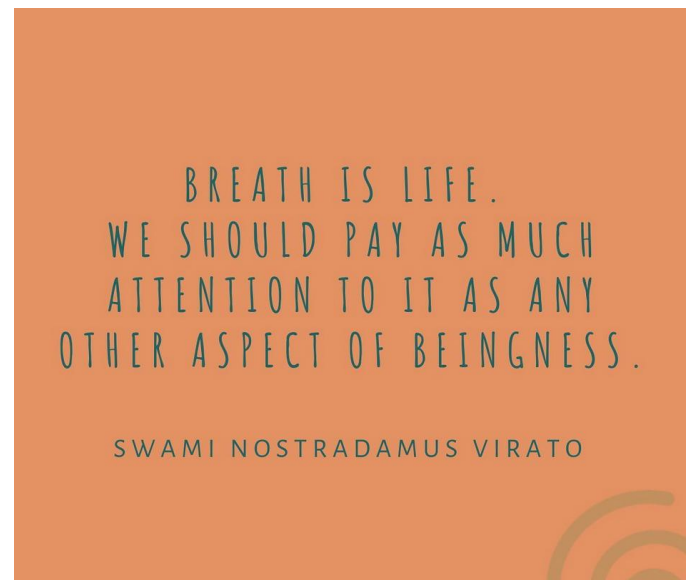


Figure 1: Breath is Life - Swami Nostradamus Virato

B. Design Idea

To combat the rising levels of stress amongst the global population we decided to join with ILU Innovations to create a breathing lamp. This lamp will be equipped with LED lights that will travel in a linear motion that will subconsciously influence

the user(s) to breathe in a specific pattern that will calm them. The light show will also be in sync with audio to give the user(s) a more transcendent experience. There will be built in sensors that are extended from the lamp that can sense breathing rate and heart rate. The user will be allowed to change the brightness of the light, its color, and the selected audio that the lamp projects.

C. Work Breakdown

The project can be broken up into three features. The ESP32 controller, Hardware Design, and System Features. The ESP32 is the microcontroller that we will be using. Its role is to convert any inputs that our sensors collect and produce the necessary outputs via LED lights, motions of the lights, and numerical data.

Our Hardware Design can be broken into a power system and a control system. The power system is responsible for distributing power as efficiently as possible to our appliance such that it can last hours at a time to limit the frequency of charging. Ideally, we wish for the lamp to run about 3-4 hours. For the control system, we will attach an IR camera that will provide an infrared imaging of the shoulders and chest. By tracking the rise and fall time of the chest, we can use computer vision that will predict the heart rate and breathing rate. We will also create a way for the audio to synchronize with the lights so they can perform in unison.

For the system features, we will use speakers attached to the lamp to provide audio for the user. A stepper motor will be used so the LED arrays can move at a specific motion. For the last feature, the user will be able to control the lamp via a mobile device. The user will be able to pick out the audio they wish to hear and the color of light to display. The user will also receive the breathing rate and heart rate via their mobile device.

D. Project Timeline

We will start off with creating a System Block Diagram, setting up the IR Camera, research miniature LED bulb and design a mounting arm which should take about 3 days to 2 weeks. We can then proceed to turn the IR image processing to detect heart and breathing rate. This should be time-consuming, which is why we allocated 3 weeks of time to work on. After designing the mounting arm, we will order a motor for linear motion and begin testing it. After that, we will order a 3D Print IR Camera Mount. When this is done, we will integrate the fully design power delivery system along with an integrated battery and a charging unit. We will then program a smooth linear movement of light and any other peripherals and Unit Testing. To finalize what we will do this semester, we will research parts of mobile application like Bluetooth connection and be ready to present our Prototype Presentation.

At the start of Spring 2026 semester, we will decide on whether we want to use an LED light array or a mechanical light bulb. We will create a list of all the necessary adjustments needed to the design. We will order parts if necessary and them to the system. We will test the whole system and continue to make improvements to anything if necessary.

E. Risk Assessment

Our project focuses on the design and development of an interactive lamp that assists users with breathing techniques by creating the illusion of a room “breathing.” Using a microcontroller to control a stepper motor, linear motion shaft, capacitive touch interface, and light source enclosed within an outer shell, the system rhythmically expands and contracts its illumination to help users synchronize their breathing with the lamp’s motion and light intensity. The capacitive touch input allows for intuitive control of lighting patterns, timing adjustments, or mode selection without the

need for mechanical switches, maintaining a seamless and modern user experience. This integration of motion, lighting, and touch control aims to promote relaxation and mindfulness through synchronized visual feedback. Because the project combines electrical, mechanical, and software components, a comprehensive risk assessment is essential to ensure safe assembly, reliable operation, and consistent performance. The following sections identify potential hazards such as mechanical pinch points, electrical shock, overheating, and software malfunction, along with mitigation strategies to maintain safety throughout the design and testing process.

F. Problem Statement Revision

Following our semester of progress on this project, our increased knowledge around the societal problem has confirmed our original design idea. However, due to minor differences in technical specifications, our finalized prototype will contain a raspberry pi instead of the initial ESP32 as expected. The overall design, features, and functions will remain the same to best serve its users and have the largest impact on the societal problem.

III. Societal Problem

A. First Semester Interpretation

Stress today can be seen in all aspects of life and deeply affects a great number of people, and we believe more easy and accessible ways to manage stress should be available. There are several different ways to relieve stress which can vary from person to person, but one universal way is the use of certain breathing patterns and techniques. Breathing techniques are easy to learn and follow, and offer a short, concise way to help gradually de-stress the body.

B. Long-term Effects of Stress

According to the ADAA, social anxiety disorder affects 15 million adults or 7.1% of the U.S. population, OCD affects 2.5 million adults or 1.2% of the U.S. population, and PTSD affects 7.7 million adults or 3.6% of the U.S. population, these are only a few anxiety disorders that are affecting our world population [2]. Stress has a big impact on our society alongside anxiety. A form of stress called job stress, this type of stress affects about one-third of workers. Healthcare expenditures are nearly 50% greater for workers who report high levels of stress, which may lead to depressive illness. Depressive illness, a common side effect of job stress, in employees is associated with nearly 10 annual sick days [9]. Estimated newly diagnosed anxiety disorders patients have increased from 31 million in 1990 to 45 million in 2019, that is about a 45% increase in just 30 years [4]. Anxiety and stress play a huge role in mental well-being and having a buildup of either will impact their lives.

Constantly being stressed with little to no management can have costly effects on work performance, mood, focus and can also disrupt a person's sleep cycle. An article on technology and its impact on modern day society investigated this topic saying "Particularly when individuals sleep with their smart phones, is this societal challenge becoming a potentially serious problem. Answering calls and text messages throughout the night only further disrupts necessary sleep. Making this a habit will ultimately cause sleep deprivation which will inevitably translate to accidents and poor work performance." [5] Sleep is crucial to maintaining a healthy body with many important biological processes occurring in REM sleep. These include repairing damaged cells, flushing out toxins in the blood, balancing hormonal levels, and most importantly reducing stress. Without sleep, stress may build up uncontrollably with little to no way of relieving it, leading to serious mental and physical implications. In an article researching the impacts of

chronic stress on the body’s physiology and psychology, it is stated that “Psychological stress can induce the acute phase response commonly associated with infections and tissue damage and increase the levels of circulating cytokines and of various biomarkers of inflammation.” [8] This means that the impacts of stress may not only be psychological but may also induce physical and chemical changes within the brain and body of those with long term exposure to chronic stress.

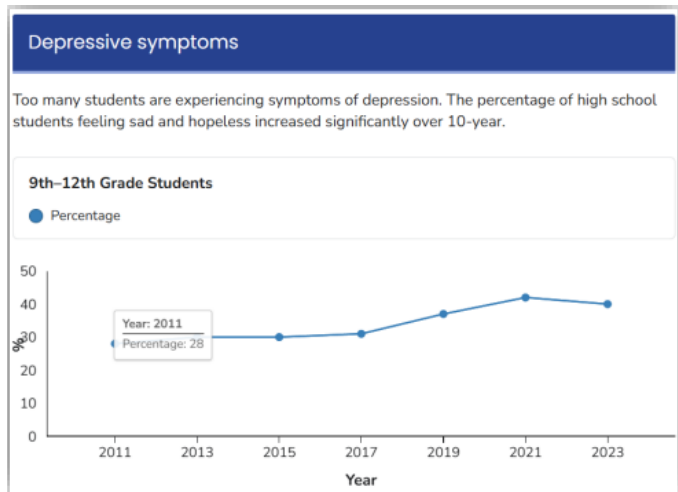


Figure 2: Depressive Symptoms in High School Students from 2011 to 2023

1) Ethical Consequences

Unmanaged stress is undoubtedly correlated with multiple disorders and afflictions, and leaving many to deal with the consequences of said stress by themselves cannot be considered a solution. Of course this would raise some ethical concerns in many. The main reason for these concerns is that if people now are struggling with chronic stress and the effects it has on their well-being, how are younger generations expected to be able to handle this issue? As seen in Figure 2, the rates of high school and early teen depression have been rising over the past decade. Of course, this has many repercussions with the younger generations experiencing academic and work-related burnout at an early age, suicide rates of teens at all-time highs, and a generally pessimistic outlook on life. It is unlikely this is the entire reason that newer

generations are struggling to integrate into society post-graduation, but it may be a cofactor in the issue. Younger generations should not be falling victim to these preventable issues and instead, underlying causes of these issues should be studied, and solutions should be offered.

Some other issues are that of stress in society causing many to lose interest in things that would benefit others such as charity work, being well mannered, offering help to strangers, and so on. Those that may have other pre-existing conditions may have an even harder time fighting or coping with their conditions with stress piling up on top of them. People that work in high-stress environments may also take a large hit if they are not able to reliably relieve the stress they incur at work. It is safe to say that chronic stress should be considered as a detriment to society and should be treated as though it is a threat to people’s well-being and dealt with accordingly.

C. Breathing as a Stress Reliever

As previously mentioned, there are many ways to relieve stress that differ from person to person. However, breathing is one thing we all share, and anyone can easily learn to do. Many studies have shown that breathing, especially slow and deep breathing, can trigger the body's natural processes to reduce stress. According to a study done on effective stress management techniques, researchers believe deep breathing is linked to the autonomic nervous system and thus can reduce systemic stress within patients. The journal states, “It is hypothesized that voluntary slow deep breathing functionally resets the autonomic nervous system through stretch-induced inhibitory signals and hyperpolarization currents propagated through both neural and non-neural tissue which synchronizes neural elements in the heart, lungs, limbic system and cortex.” [10] If deep breathing is closely connected to the body’s autonomic nervous system, it can be used as an effective way to regulate and

mitigate stress by resetting potential stressors within said nervous system. In the same study, the results made it clear that this is in fact the case and may also be used to help patients with pre-existing conditions. The journal's results were that “it has been used to influence autonomic functions in patients with essential hypertension and thus reduce it, in the management of male adolescent aggressive behavior, in long term prophylaxis of migraine, in stress related to dental visits.” [10]

Breathing practices have been proven to help with stress and anxiety. Studies show breathing practices reduce stress, anxiety, and depression while improving sleep and emotional regulation. Providing a safe, effective, low-cost tool that benefits mental health aligns with the duty to do good. Breathing techniques can be practiced independently without constant reliance on healthcare providers. Unlike medications, breathwork typically carries very few side effects when practiced responsibly. This minimizes harm while still offering therapeutic benefits, which makes it ethically favorable as a stress-reduction strategy. Breathing practices are low-cost and can be taught widely in schools, workplaces, and communities. This promotes fairness by giving more people access to mental health support, especially those with limited access to medical care. Articles like the MDPI review highlight both successes and limitations, encouraging evidence-based application. Acknowledging limits prevents overclaiming and builds trust between practitioners, researchers, and the public. [5] Modern society is facing a growth of mental disorders, with stress emerging as a critical factor affecting mental well-being. In recent years, breathwork has gained public and scientific recognition as a promising approach for enhancing psychological health. “Accumulating evidence suggests breathwork may serve as both a preventive and adjunctive therapy for chronic stress, anxiety, and depression, given its potential to target

key risk factors and produce clinically relevant outcomes.” [3]

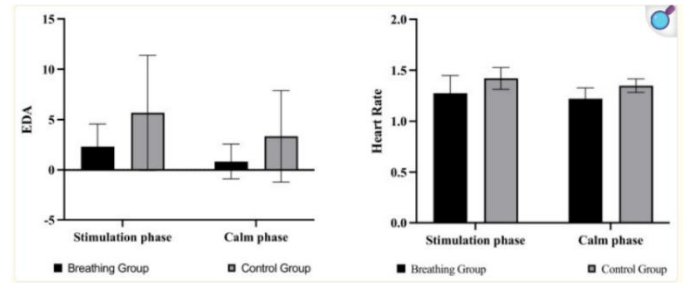


Figure 3: EDA values and Heart Rates for respiratory and control groups

Diaphragmatic breathing seems to be, in many cases, a reliable way of reducing the effects of chronic stress on the body. As seen in Figure 3, a study was conducted in which two groups were put into stress-inducing situations in which their heart rate would naturally begin to increase. One group was the control group in which they were told to react how they normally would in any stressful situation. The other group was told to employ common breathing patterns to calm themselves during the situation. The group which employed the breathing patterns had a noticeably lower average heart rate than the control group both during the stimulation, and after. This shows that diaphragmatic breathing methods can not only reduce the amount of stress imparted onto someone during a stressful event but also may decrease the negative effects that stress can have on the body. We believe this will be a key part of our design to solve these issues.

D. Proposed Solution

Our solution to this issue is to design and create a light-based system that patients can synchronize their breathing within order to stimulate the body's autonomic nervous system and start the natural processes that relieve stress. This device will be based on a microcontroller that will control the movement of a motor which is physically connected to an LED light bulb in such a way that it will induce linear vertical motion. This motion of the

light bulb will shine through an outer shell and project the different shadows cast by the shell to give the entire room a calming and soothing atmosphere. The vertical motion will also be synced to common diaphragmatic breathing patterns so that when a user sees the motion of the light, they are naturally inclined to sync their breath with the motion of the light. In addition to the light, we plan to add speakers to the system which will be able to play music or sounds that match the tone and rhythm of the light's motion, to further create a calming atmosphere.

We believe the system should also be interactive in some way, syncing with the heart and breathing rates of those that choose to interact with the sculpture. To achieve this, we will use Infrared sensors mounted to an external pedestal that when a person is standing in range of, will detect the heat from their body that pulses with their heartbeat, and process the IR patterns to estimate an average heart rate. The system will then adjust the lights, motion, sounds, and other features to a rhythm that is slower than the person's heart rate to help slow their breathing and heart rate. To do this, we would first need to test to see if an IR camera or sensor can reliably pick up a person's heartbeat from a few feet away. If the IR camera shows a pulsating pattern on its video feed, we would then need to program and train a small AI model, or use existing models that can recognize the patterns, and synthesize an accurate estimate of the person's heartbeat.

All features should also be optimized to reduce battery and power usage for a long battery life, and efficient discharging and recharging. Any code that is written and used should be as efficient as possible in this right to further increase battery life and reduce any power losses from redundant code execution.

If time permits, an additional piece of interaction could be added via a phone or tablet application that can be connected to the device using Bluetooth

technology. Using the application would allow the customization of colors, motion rate, music, and sounds that the system will display.

IV. Design Idea

A. Design Philosophy

To address the societal issue of increasing levels of stress from several sources, our team will construct a light-based system that will simulate and mimic common breathing patterns to help reduce stress. Deep, methodical breathing has been proven to help alleviate stress and is something everyone can incorporate into their daily lives with ease. Our vision is that the system can be used in public gathering areas as an expressive work of art, in which people can view and interact with to help passively calm their breathing, and reduce their heart rate. In theory this will help oxygenate the person's blood, activate the body's natural repair processes, and ultimately reduce stress. We believe our idea is innovative and unique because it will passively help people to alleviate stress with very little to no conscious effort or involvement. Other ways of relieving stress may be very effective; however, many require dedication and focus to effectively relieve stress. With the breathing light, simply being around the light and in the created atmosphere will help to calm breathing and reduce stress.

B. Specific Design Components

1) Technical requirements

The main component of the system which ties everything together will, of course, be the microprocessor. This microprocessor, which is the Raspberry Pi, will connect all the other components together and must be able to effectively and efficiently process the signals required to operate all functions of the system. The Raspberry Pi should be robust enough to do all the necessary signal processing and calculations, but also efficient to save power and increase battery life. Below is the initial Punch List that contains all the desired

features of the system, and a measurable metric that must be met to achieve the desired feature.

Table 1: Fall 2025 Punch List

Feature	Measurable Metric
Vertical Motion of LEDs that mimic breathing	Smooth linear motion that can be controlled with accuracy and precision within 15 msec
Sensor that will detect heartrate and adjust the rate of other features	Within 5% margin and synchronization with other features to be within 15 msec
Sensor that can detect breathing rate and adjust features accordingly	Within 10% margin and synchronization with other features within 15 msec
Audio that can be synced to the rate of the light's motion	Audio that matches bpm to the linear rate of motion of the light within 0.5 bpm
Efficient and rechargeable battery	Battery life of at least 6 hours, and fully charging within 2 hours
Connection and control via mobile application	Users can select a variation of light or audio via Bluetooth connection.

To create the vertical motion, either a linear actuator or a mechanical motor connected to a crankshaft type assembly will be used. Pulse Width Modulation can be employed by the microprocessor to control the rate at which the light will move, creating accurate and precise movements. To detect an individual's heart and breathing rate, two different sensors will be used. An Infrared (IR) camera will detect the pulsating heat given off by a person's beating heart. And to detect breathing, a depth-sensing camera system which will measure the movement of the individual's chest. For wireless connectivity, the microprocessor we have chosen will include Wi-Fi 6/Bluetooth capabilities so all that is needed is to program and integrate wireless connectivity and a paired application that can be

used for testing. To include audio playback, an external audio amplifier will be required as our chosen microprocessor does not have an on-board amplifier. This will be connected to a speaker or speakers for total coverage of the entire space. An interface between the light's motion and the audio playback system must be programmed to achieve synchronization between both features. Of course, all these modules and features will require power to work, so an external 1000 mAh rechargeable battery will be connected to the microprocessor and each module. This rechargeable battery must also be capable of being recharged via standard 110V wall outlet and should charge reasonably fast.

To interface with the microprocessor and write programs, an Integrated Development Environment (IDE) must be used while the microprocessor is connected to a computer. For the microprocessor we will be using, the Arduino IDE must be used to write code to the processor. As the Arduino IDE contains everything needed to develop, test, and debug programs, no other software like compilers or interpreters will be needed for the microprocessor.

2) Skill Requirements

Of course, developing, testing, debugging, and assembling a system like this will require a diverse set of skills to effectively complete the project. Many of the skills required have been learned, applied, and refined in previous courses and will be applied to the best of our knowledge. Nicholas Her can assist with Java or C language to code the microcontroller. He can also use his handyman skills to create and design our proposed shell. He's had previous exposure to sensors, which can assist with our sensors that are needed to detect heart and breathing rate. Michael will aid in circuitry by wiring the microcontroller with other components and help with the schematic of the circuit. Eric has experience with programming and debugging microcontrollers in the C language which will prove valuable in this project as there will likely be a lot of programming and debugging required. He also

has experience with designing and building circuits, so he can help with any circuit-related tasks if needed. Paul also has extensive practice with coding in Java and C, especially focused on resolving issues and increasing code readability. Additionally, he has previous experience with designing and producing mechanical systems, which will be a substantial portion of this project.

V. Funding

As we have been contracted by a sponsor to develop this project and will be closely working with them to achieve their envisioned design, all funding will come directly from our sponsor, Ilu Innovations Inc. Ilu Innovations has generously given us an initial amount of \$500 to spend on all necessary parts and components required to develop and complete the system. An additional \$500 can be requested if the initial amount is used in its entirety, however we believe the initial \$500 will likely be sufficient to purchase all components. In addition, Micah Kearns, the sponsor representative who we are closely working with, has offered to supply his own parts and materials for the system's outer shell and housing.

A. Distribution of Funds

The initial amount of \$500 will be allotted to Nicholas Her, who will oversee keeping track of all purchased items and invoices. He will keep track of who has purchased what items and reimburse them accordingly. Items purchased will also be logged into a shared Google Sheet with price, quantity, and any related invoices or receipts attached.

VI. Work Breakdown Schedule

A. Project Tasks

The development of this system will need to begin with considerable planning and many decisions around system specifications and functions. In the first few weeks, all members will be working on defining the whole system specs. This includes key features, physical components, software features and overall goals. Then over a few days, our group will work on creating the system's block diagram and the data flow between components. This will help us to get a broader view of each part of the system and the steps needed to get them all to work together properly. The next step will be to select specific components and make purchases. This will be primarily done by Nicholas and Eric to simplify the funds handling. Once all components are received, we will test and verify that they can all communicate with the Raspberry Pi and function in the way we intend to use them. We will also test to confirm efficient power consumption to maintain targeted battery life specs.

Once the planning and selection process is completed, our group will begin to work on designing the hardware and power system. Michael will primarily focus on designing the power distribution and circuitry, including the components' power requirements and battery/AC input. Next, Michael will physically integrate the battery, circuit, and voltage regulation, to then test and confirm expected performance and battery life.

With the power system in place, our group will begin working on the control system and peripheral components. Paul and Michael will connect the Raspberry Pi to our selected actuator/motor for physical movement of our light array. This will be our first major feature, which will be to stimulate breathing through the movement and changing of the lights. Paul and Eric will implement GPIO and PWM controls for the LEDS and motion driver for

the realistic and adjustable breathing effects. Meanwhile, Nicholas and Michael will be designing connection routing for the IR sensor and audio amplifier, to ensure clean cable management as well as clean data and sound. Paul will also work on building and connecting the LED array circuitry. Michael will be leading the progress on wiring and integrating the IR sensor to ensure stable power and proper data connectivity.

Once all hardware is integrated and properly configured, our team will be able to fully focus on implementing the key system features. Eric will program the LED patterns that will stimulate the breathing through rhythmic light motion. Paul will work on controlling the motor/actuator and programming the adjustable speed and rotation to further enhance the natural breathing effects. For biofeedback, Nicholas, Paul, and Eric will be responsible for integrating the sensors that will detect heart rate and breathing rate using the IR camera, conductivity sensors, and/or haptics. Michael will focus specifically on the IR camera and handling the data it collects. Together, Eric and Paul will write the logic for the system that will, in real time, adjust the movement speeds, light colors and brightness, and the sound's rhythm, pitch and volume.

After all hardware and software features are implemented and properly working, the team will expand functionality through Bluetooth communication, a controllable mobile application, and additional options for the lights and sound configurations according to the input breathing/heart rates.

Toward the end of this process, our group may additionally create pre-defined programs or themes for users to experience, without needing the IR camera or other included sensors.

VII. Project Timeline and Milestones

A. Major Milestones

In the previous section, our project was split into smaller tasks that must be completed to complete the whole project. To stay on track for completion and use our time efficiently and effectively, major milestones have been placed at strategic dates. The First major milestone to be reached is the integration of the IR Camera into the microcontroller and Image Processing successfully giving an accurate Heart and Breathing Rate. This Milestone must be met by Saturday, November 8th as it will allow us to implement and test the physical motion of the light. Completion of this milestone will allow us to work on tasks towards the next major milestone, which is the integration of all other peripherals and the power delivery system. This must be completed by Thursday, December 4th to have the prototypes working for the presentation on Friday, December 5th.

B. Second Semester

Our goal by the end of this current semester is to have 2 rough, working prototypes of our system. One will use a motor to drive a light bulb attached to the end of a rod in a vertical motion, and the other will use a custom LED array programmed to mimic the vertical motion of the motor driven system. Our sponsor will then choose which design they like best, where we will then use the next semester to refine that design, and implement new features to the design. Both systems will use identical IR Camera sensors to detect Breathing and Heart Rate and power delivery systems, with some minor changes. The main difference will be the method in which the movement of the light is done. As we are unsure which design will be chosen for the second semester, the Gantt Chart for Spring 2026 has been filled out with generalized tasks that

can be applied to either design, and will be refined once a design has been chosen.

C. Documentation

Documentation will continue to be written and added to this report to ensure thorough understanding of how our systems were designed and implemented. Weekly reports will also be written and turned in to keep Professor Levine up to date on our progress towards completion. All necessary assignments will be added to the team Microsoft Planner with appropriate due dates, including Second Semester assignments for organization and proper time management. This includes all weekly activity reports, team member evaluations, risk management assessments, deliverables and presentations, and technical showcases.

VIII. Risk Assessment

A. Overview

This section identifies, analyzes, and ranks the potential risks associated with the development and testing of the breathing lamp project. The system combines electrical, mechanical, and software components, including a microcontroller, stepper motor, linear motion shaft, capacitive touch interface, and light source housed within a wooden outer shell. The lamp is designed to simulate a rhythmic “breathing” effect through synchronized motion and illumination to promote relaxation and controlled breathing. Because the project involves precision motor control, electrical wiring, heat generation, and sensitive touch electronics, it presents multiple potential hazards during assembly, integration, and testing. This risk assessment outlines each identified risk, evaluates its potential impact, and presents corresponding mitigation strategies to ensure that the device can be safely constructed and tested in a laboratory environment.

B. Identified Risks and Mitigations

1) Mechanical Risks

During assembly and testing, the stepper motor and linear motion shaft present potential hazards due to pinching and entanglement. Accidental contact with the moving mechanism could cause minor injuries or component damage if the system activates unexpectedly. To minimize these risks, the team will ensure that the motion assembly remains powered off during adjustments and only operates the system when all components are securely fastened. Protective gloves will be worn when handling the linear shaft, and all mechanical elements will be mounted on a stable base to prevent shifting or tipping during testing.

2) Electrical Risks

The project’s electrical system involves the microcontroller, motor driver, and power supply, all of which pose risks of shock, short circuits, or component burnout during prototyping. Loose wiring or incorrect connections could lead to arcing or overcurrent conditions. To mitigate these hazards, the team will double-check all wiring against circuit schematics before powering the system. A current-limited bench power supply and fuses will be used to protect sensitive components. Members will avoid working on live circuits, use insulated tools, and follow standard lab safety procedures for electrical work.

3) Thermal and Fire Risks

The light source and electronic components may generate heat during extended testing. Without proper ventilation, this could lead to overheating or even a small fire risk, especially with the wooden outer shell. To address this, the team will perform power-on tests in supervised lab settings and limit test durations. Adequate airflow will be maintained through vent openings, and no flammable materials will be placed near the lamp during operation. Fire-retardant coatings or heat-resistant inserts will be used when necessary, and a fire extinguisher will be kept nearby during all powered tests.

4) Human and Operational Risks

Risks during project construction include improper handling of tools, accidental contact with powered components, and injuries from sharp or unfinished materials. To reduce these risks, the team will use proper personal protective equipment (PPE), including safety glasses and gloves, while cutting or sanding the wooden shell. Only one person will work on live circuits at a time, and all power will be disconnected before adjustments. The team will maintain an organized workspace to avoid tripping hazards and follow all laboratory safety rules and supervision requirements.

5) Touch Interface Risks

The addition of a capacitive touch interface and touchscreen introduces new electrical and control concerns. During prototyping, improper grounding or insulation could cause electrostatic discharge (ESD), potentially damaging the microcontroller or the touchscreen's sensor circuits. Furthermore, software errors in touch calibration or sensitivity settings could cause unintentional system responses or motor activation. To mitigate these risks, the touchscreen will be integrated with proper ESD protection components such as transient-voltage suppression (TVS) diodes and grounding pads. Testing will be conducted using antistatic wrist straps and mats. Calibration routines will be validated in software to ensure that accidental touches or false readings do not trigger motion or lighting changes. Additionally, all team members will handle the screen by its edges and avoid applying excessive pressure during installation to prevent cracking or delamination.

6) Environmental Risks

Dust from sanding, debris, or moisture exposure during assembly could interfere with electrical components or moving parts. Additionally, vibrations during testing could cause screws to loosen over time. To mitigate these risks, all components will be kept in clean, dry areas, and work surfaces will be regularly cleared of debris. The team will check fasteners after each test and ensure that all wiring and parts are properly secured to withstand operational vibrations.

7) Software and Control Risks

Improper coding or motor control parameters could cause **unexpected motion** of the stepper motor or excessive current draw. These errors could damage components or create a mechanical hazard for the builders during debugging. To prevent this, the team will test all motor control codes with limited current and reduced speed settings before full operation. Software will be reviewed and simulated prior to

uploading, and an emergency stop function will be implemented in both software and hardware. Consistent documentation and version control will be maintained to track and revert firmware changes safely.

C. Risk Chart

Risk	Likelihood (1 – 5)	Risk Level	Mitigation Strategy
Pinching or entanglement from moving shaft	3	Medium	Power off during adjustments; use stable mounting
Electrical Shock or short circuit	2	High	Use correct tools, fuses, and current limited power supplies; verify wiring before powering system
Overheating of light source or electronic components	3	High	Ensure ventilation, use heat resistant materials, and limit test durations.
Injury from tool use or assembly processes	3	Medium	Follow proper tool handling procedures.
Electrostatic discharge (ESD) damaging capacitive touch interface	3	Medium	Use ESD safe workstation, grounding straps, and TVS diodes on input lines.
Software malfunction causing unintended motion or light output	3	Medium	Conduct simulation and staged testing; implement emergency stop logic and

			watchdog timer.
Component loosening due to vibration or repeated testing	2	Medium	Use thread-locking compounds, lock washers, and inspect hardware regularly.
Dust or moisture interfering with electrical components	2	Medium	Keep workspace clean and dry; seal openings and coat exposed circuits.
Power supply mismatch or voltage instability	2	High	Confirm correct voltage/current ratings before operation; add voltage regulators and fuses.

D. Compliance and Safety Procedures

The team will comply with all University, state, and federal safety regulations while constructing and testing the breathing lamp prototype. All work will be conducted in designated laboratory environments under the supervision of course instructors or lab staff when required. Standard laboratory practices will be followed, including the use of personal protective equipment (PPE) such as safety glasses, gloves, and closed-toe shoes during fabrication and assembly. Electrical work will be performed only when the system is de-energized, and components will be properly grounded to prevent electric shock or damage. During soldering or cutting operations, adequate ventilation will be maintained to minimize exposure to fumes and dust. To reduce the risk of electrostatic discharge while handling the capacitive touch interface, team members will use grounded antistatic wrist straps and mats. In accordance with university health and safety guidelines, team

members will maintain a clean workspace, store tools and components safely, and observe proper hygiene and distancing practices as required. These procedures collectively ensure that the project is completed in a safe, controlled, and compliant manner.

E. Summary

Through this risk assessment, the team has identified and analyzed the primary hazards associated with the construction and testing of the breathing lamp prototype. These include mechanical hazards from moving components, electrical and thermal risks from powered systems, software control errors, and electrostatic discharge concerns related to the capacitive touch interface. Each identified risk was evaluated for its likelihood and severity, and mitigation strategies were developed to minimize potential harm to team members and damage to components. By implementing these safety measures such as proper grounding, protective enclosures, fire prevention practices, and electrostatic discharge protection, the team ensures a safe and reliable development process. The proactive approach taken throughout this assessment not only supports successful prototype completion but also establishes strong engineering and safety practices for future project phases.

IX. Device Test Plan

A. Linear Actuator Tests

For the linear actuator portion of the project, which consists of the motor, shaft, and internal lead screw assembly, we plan to conduct functional, performance, and durability testing to ensure reliable operation. The actuator will first be tested for basic functionality by running it through full extension and retraction to verify smooth motion without binding or mechanical interference. Speed testing will be performed by measuring the time required to travel a known linear distance and calculating linear speed, with trials conducted at multiple command input levels to confirm that motor speed corresponds accurately to the values being sent by the control system. Command accuracy will also be verified by comparing transmitted control signals to the received values at the motor driver and confirming consistent mechanical output. Load testing will be conducted under expected operating conditions to ensure the motor and lead screw maintain performance without stalling or excessive slowdown. Finally, a durability test of at least 100 full extension–retraction cycles will be executed while monitoring travel time and overall behavior to confirm repeatability and identify any performance degradation. These tests together will validate mechanical reliability, speed accuracy, signal integrity, and long-term performance of the actuator assembly.

B. Heartrate Detection Tests

To test the functionality of our heartrate detectors and verify that they can read accurate heart rates of subjects, the system will be exposed to different kinds of environments and lighting conditions. Currently lighting and movement are the two biggest issues the detection system faces, and optimizations will be implemented to combat them. Testing of the heart rate detector may include testing the device at different times of day and

different weather conditions or artificially changing the ambient light inside the testing area. Movement is also a critical issue with detection; however, it must be violent and sudden to completely lose track. Subtle movement is much less likely to cause a complete loss of tracking and has been accounted for in the code. There is little that can be done to combat the issue of movement without taking a considerable amount of the projects budget. A test to confirm the system can handle more subtle movement will be included in the test plan, however. This test will mainly focus on testing the accuracy of the heart rate detection while a subject is moving slowly in the field of view and stationery with some small adjustments like tilting of the head. Additionally, testing the actual accuracy and precision of the heart rate detectors will be necessary. This can be done by comparing the results of the detector with a piece of equipment that is proven to be accurate at reading heart rate. This can be a simple wearable heart rate monitor such as a smart watch or fitness tracker, or something more sophisticated like an ECG. Testing different heart rate values should also be included by incorporating different subjects as well as subjects that perform some exercise movements and then immediately interact with the detectors.

C. Linear Actuator Test Results

The linear actuator assembly demonstrated satisfactory performance during testing and met the current project requirements. The actuator consistently achieved full extension and retraction with smooth motion and no observable binding, mechanical interference, or abnormal noise. Manual command values were received correctly by the system, confirming proper signal transmission, although additional testing with the IR camera is still required to evaluate live data integration. Load testing was performed using the lightbulb, which represents the intended operating load, and the actuator operated without stalling, overheating, or any noticeable loss in speed or torque. Each test was

conducted for a minimum of 10 minutes to assess consistency of operation. During testing, one bolt securing the gear loosened due to vibration; this issue was corrected by applying Loctite to prevent further movement. Overall, the actuator performed reliably under the tested conditions, with only minor additional validation remaining for the complete system.

D. Heartrate Detection Test Results

The results of the Heart Rate Detection System Tests were a pass on both the Heart Rate Movement Accuracy Tests and the Heart Rate Lighting Tests, meaning that the system will work accurately and reliably in most conditions. The system performed very well during the movement test, deleting any frames that detected the subject moving rapidly within the detection window, which could cause inaccurate readings. This was an intentional implementation because rapid movement of a subject will result in blurred frames that can drastically change the readings from the system. Because of this, a software filter was implemented that deletes any frames that detect rapid movement of the subject, resulting in a more stable and accurate overall reading. This feature was tested by having the subject move within the detection window at different speeds and seeing how the heart rate would vary and if it stayed accurate to the actual subject's heart rate. To get a solid control of the subject's heart rate, a fitness monitor that is strapped to the chest of the subject was used, which would then be compared to the readings from the Heart Rate Detection system. This control was used on both the Movement and Lighting tests.

For the results of the lighting test, the system performed very well in all kinds of indoor settings. Only the darkest indoor lighting caused issues with the system, however this can be solved by switching the mode of the camera to NIR and using the IR detection algorithm. The system also struggled with

very bright, direct sunlight as it would blow out the camera resulting in a very inaccurate heart rate. The result of the test is still a pass, however as the system performed very well in all other lighting conditions and is intended to be used in mild conditions for the best possible heart rate readings. Some future adjustments or customizations could be made to get a robust reading in more extreme conditions such as those very dark or very bright settings.

E. Sound System Test Results

The results of the sound system came back well. When I attached the speakers to the breadboard that was connected to the MP3 module and the ESP32 connector, the level of quality that the audio displayed was pretty good. There was hardly any muffling which shows how signal integrity was not lost on the embedded side of this system. When it came to the volume of the speakers, I found it to be quite satisfactory for what we were trying to achieve. The level of volume remained the same every time we tested it using different audio files. When it came to determining how accurate the system would find the correct audio file depending on the dummy heartrate that was given, it was fairly accurate overall. When I assigned 100 dummy heartrate values, the program was able to output the correct corresponding audio file at 98 percent accurate. With this level of accuracy, the audio files selection should be good enough when its time to intertwine it with the other moving components. Lastly, I tested how quickly the audio would play once the heart rate has been captured. After repeated testing, I found out that it takes no longer than 2 seconds for the audio to play once a feed in a dummy value to act as the heart rate value.

X. Marketability Forecast

For the marketability of this product, we wanted to pinpoint the traits that our consumer would have. The traits would include people who are interested in relieving any stress/anxiety from things such as school, work, or everyday life. Also, people who are interested in a display piece that will offer a more aesthetic feel to their living environment. We concluded that these were the traits our consumer would want and that the lamps would be in places such as a private office for a person to enjoy or can be placed in public offices for numerous of people to experience. Since our product comes with a function that can capture heart rate and translate it into breathing rate, having it placed in a doctor's office or hospital would be appropriate. Due to the soothing music that comes from the speakers of the lamp and the visual display, the lamp can be placed in libraries and even in living spaces such as bedroom, living room, etc.

Our competitors when it comes to reducing people's stress/anxiety are meditation apps, white noise machines, and light therapy lamps. All three are good solutions to ease someone's stress but we decided to create a product that would essentially combine all three together. When it comes to our competitors that are lamps themselves, we believe that our product is best served by the industry due to its variety of functions. The breathing lamps such as the "Light Therapy Lamp" and the "Voraiya Lamp" are great lamps that consumers are already familiar with.

What makes our product different from other breathing lamps is that it can track the heartrate. We have a built-in IR camera that can capture the heartrate based on the color changes of the user's forehead. We also have a sound system that can play soothing music that correlates with the users' heart rate. When it comes to aesthetics, our lamp and our competitors have visually pleasing, but we have a hood for the lamp that when it covers the

lightbulb, patterns get displayed to the walls of the room that matches the breathing rate of the user.

One of the strengths of our product is its ability to interact with the user. Based on the data that's been captured from the user, the sound/color output, and the visual display will defer. This allows the buyer to have unique experience with the lamp compared to other lamps on the market. When detecting heartrate, tools such as smartwatches, chest straps, and armbands must make physical touch with the user to capture a heart rate. With our rPPG technology, the product can detect heart rate using an IR camera leading to zero physical connection.

XI. Conclusion

A. Societal Problem

Stress is a pervasive issue that affects nearly every individual at some point in life, and unmanaged stress has significant consequences on both mental and physical health. As studies show, chronic stress contributes to anxiety disorders, sleep disruption, and decreased productivity, while also driving up healthcare costs for society. Since avoiding stress altogether is unrealistic, effective management strategies are essential. Breathing based approaches offer an accessible, low cost, and scientifically supported method to reduce stress and improve wellbeing. Addressing stress management not only benefits individuals but also strengthens communities by promoting healthier lifestyles, reducing workplace strains, and improving overall quality of life.

B. Design Idea

We believe that what we designed will be effective due to how the user doesn't have to be conscious of it for it to work. In most cases when it comes to dealing with stress, the user must give their full, undivided attention to the action taking place. With the breathing lamp, the user(s) can do any activity that they were doing such as reading or studying and will still receive the full benefits for the breathing lamp. Also, the breathing lamp mobile can be treated to not just one user but multiple people in a large setting. It can be used in any enclosed space such as a library, hall, or classroom.

C. Work Breakdown

The development and implementation of this system has many different features and functions that have been defined in the Design Idea section. To effectively implement all these features, the system has been broken down into phases in which several smaller tasks will be completed, eventually leading to the completion of that phase and features. The

first phase will be part acquisition and testing. The focus of this phase is to research and order all the parts that are needed to build the features laid out in earlier sections. Ordering will primarily be done by Nicholas and Eric to avoid unnecessary transactions and confusion, and parts will be distributed accordingly. Once parts arrive, testing can begin to ensure the conceptual plan of each feature will work as intended and useful data can be gathered.

Once testing has been completed, we can move to the next phase of the project which is power and hardware design. This phase will be used to design block diagrams and data flow models of the power systems and the hardware features. These models can then be broken down into programming tasks and circuit designs which will be distributed to members for completion.

After the power system has been designed and implemented, the control systems and peripheral phase will begin. More components for peripheral features will be ordered during this phase and the circuit design and data flow models for the peripheral features will be drafted. Programming and integration of systems into the microcontroller will also begin in this phase, however, complete integration of features will not be worked on until the next phase.

That phase will be the system integration phase where all key features will be integrated into the microcontroller. This phase will also accompany many tests to ensure all features are working as intended, but hopefully any bugs and problems will be handled within each phase to reduce overall issues in the end.

D. Project Timeline

The project's timeline is clearly structured around two critical milestones before the semester ends, followed by iterative refinement next semester. Our team has a strong emphasis on organization, documentation, and communication, ensuring

consistent progress tracking and accountability. The planned structure reflects a methodical and professional engineering workflow prioritizing functional validation first, then moving toward optimization and feature expansion.

E. Risk Assessment

The development of the breathing lamp project presented several potential risks related to its electrical, mechanical, and software components. Through a detailed risk assessment, the team identified possible hazards such as mechanical pinching, electrical shock, overheating, and electrostatic discharge associated with the capacitive touch interface. Each risk was carefully analyzed, and mitigation strategies were established to ensure safe assembly, testing, and operation of the system. By implementing protective enclosures, proper electrical insulation, current-limiting devices, and strict laboratory safety procedures, the team has minimized the likelihood of accidents and component failures. In addition, thorough software validation and the inclusion of emergency stop functions enhance the overall reliability of the prototype. The team's proactive approach to identifying and managing risks ensures a safer development process and establishes a solid foundation for future iterations and improvements of the device.

F. Device Test Plan

Short summary of test plan

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Glossary

ADA (Americans with Disabilities Act) - a law passed in 1990 that prohibits companies from discriminating against people with disabilities such as employment.

REM Sleep (Rapid Eye Movement Sleep) - a unique phase of sleep-in mammals (including humans) and birds, characterized by random rapid movement of the eyes, accompanied by low muscle tone throughout the body, and the propensity of the sleeper to dream vividly.

Cytokine - proteins created by cells to regulate immune systems and other cellular activities.

Biomarker - measurable indicator of some biological state or condition.

Autonomic Nervous System - a branch of the peripheral nervous system that controls involuntary bodily functions, such as heart rate, digestion, and breathing.

Hyperpolarization Current - inward cationic currents in excitable cells like neurons and heart cells that activate during hyperpolarization (a decrease in membrane potential). These currents, which carry both Na^+ and K^+ ions, are important for setting resting potentials, controlling neuronal integration, and contributing to cardiac and neuronal pacemaker activity.

Appendix A. Hardware

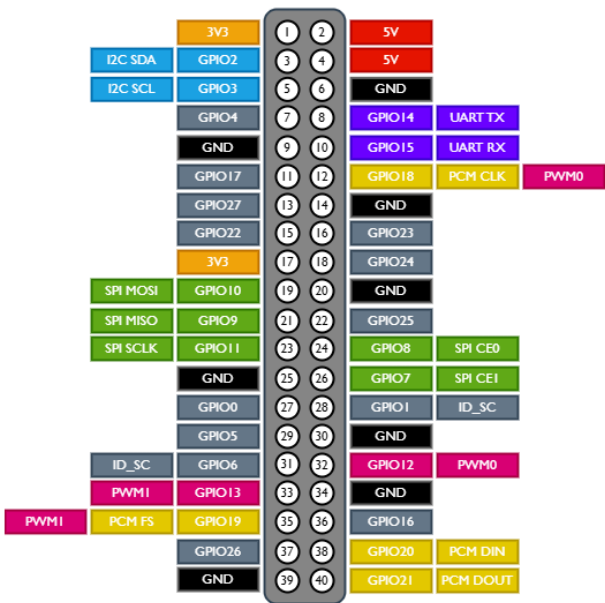


Figure A1: Raspberry Pi 5 GPIO Pinout

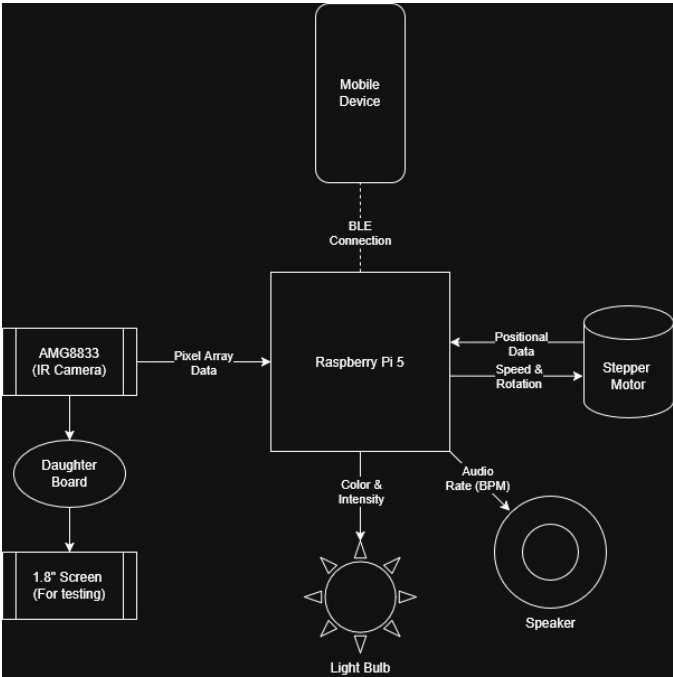


Figure A2: Hardware Block Diagram

Appendix B. Software

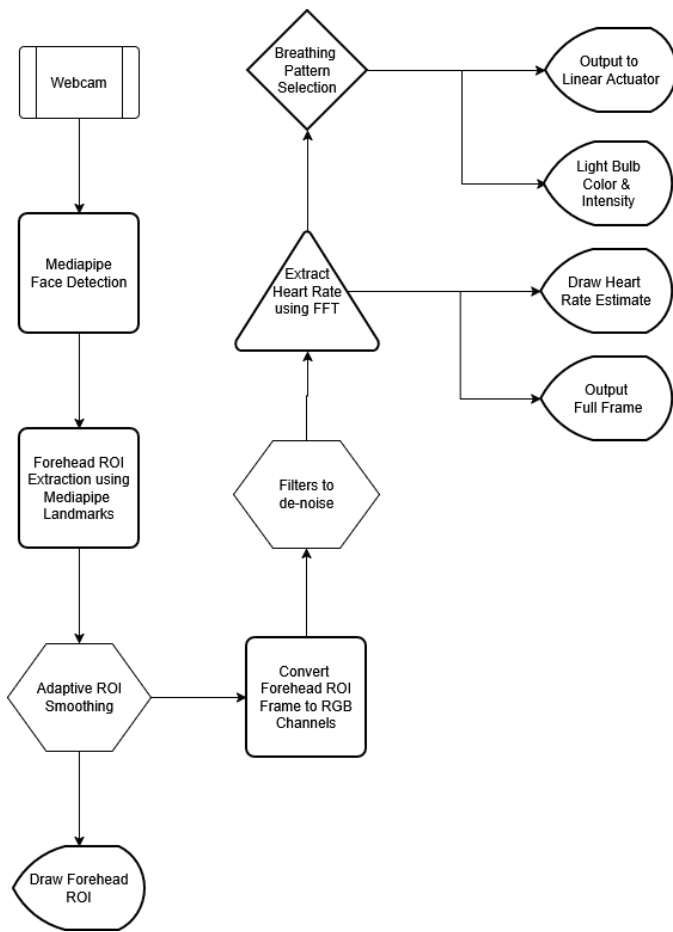


Figure B1: Software Data Flow Diagram

Appendix C. Mechanical Aspects

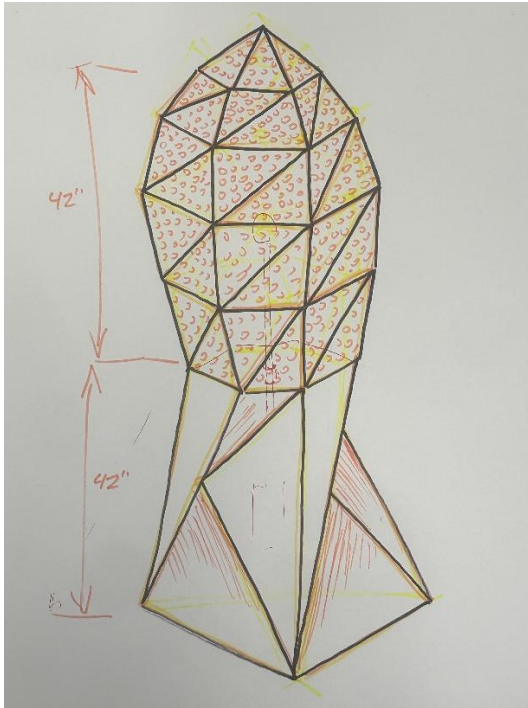


Figure C1: Proposed Outer Shell Design

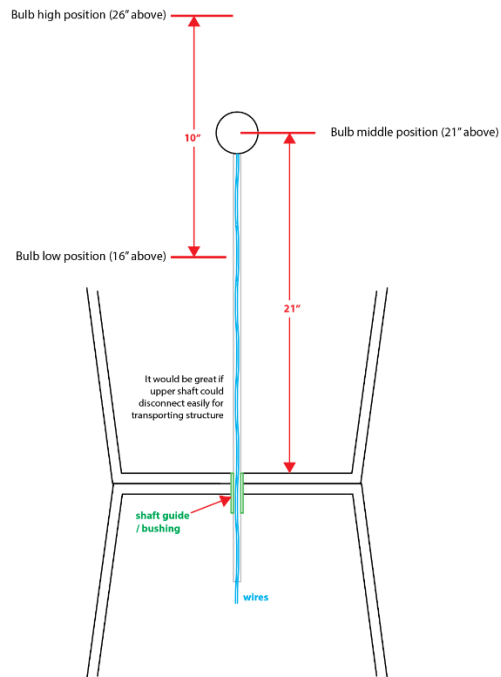


Figure C2: Shell Design for Shaft Guide

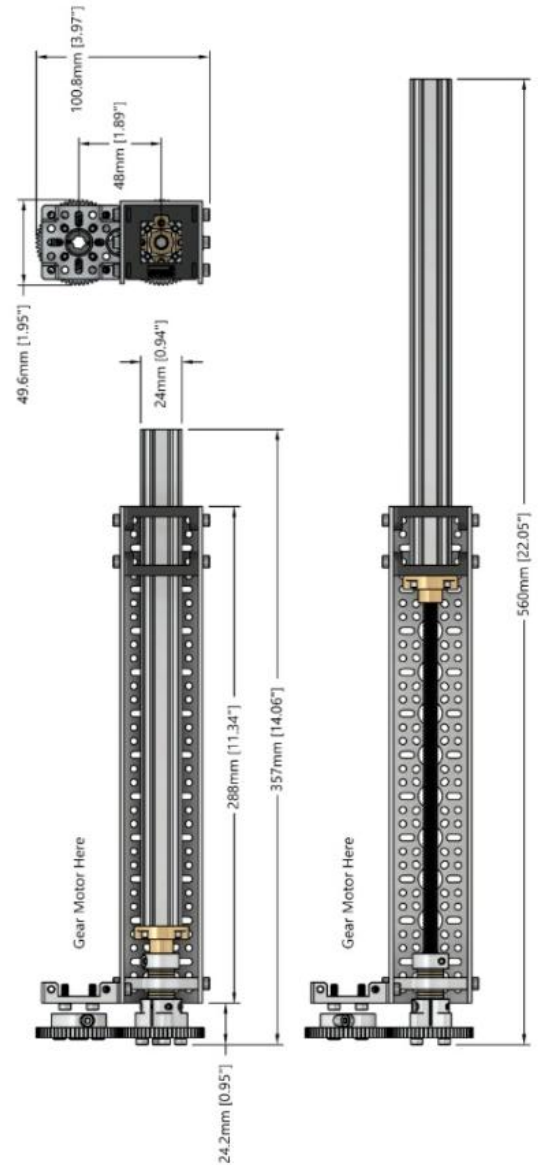


Figure C3: Linear Actuator Design & Dimensions

Appendix D. Work Breakdown Structure

Table D1: Work Breakdown Structure Overview

Level 1	Level 2	Level 3
Raspberry Pi Microcontroller		
	Define whole system specs.	
		Create system block diagram and data flow
		Select components (motor, sound amplifier, IR sensor, battery, etc.)
		Verify electrical and communication compatibility
Hardware Design		
	Power System	
		Design power distribution (battery, AC input)
		Integrate Battery and charging circuit
		Implement voltage regulation and protection for circuits
		Power budget and testing
	Control System	
		Connect Raspberry Pi to actuator/motor

		Implement GPIO/PWM controls for LEDs and motion driver
		Design signal routing for IR sensor and audio amplifier
	Peripheral Components	
		LED array control circuitry
		Audio Amplifier wiring and speaker mounting
		IR Sensor power and data connections
System Functions and Key Features		
	Motion of LEDs that stimulate breathing	
		Custom LED Array
		Stepper Motor Control
	Sensor that will detect heartrate and adjust other functions	
		IR Camera, haptics, or conductivity sensors
	Sensor that can detect breathing rate and adjust features accordingly	

		IR Camera
	Audio that can be synced to the rate of the light's motion	
		Internal speaker connected to microcontroller
		Pre- programmed or adjustable rhythms, pitches, and sounds
	Long Lasting and Rechargeable	
		Efficient Battery and circuit design
	Connection and control via mobile application	
		React Front End
		Java Back End
	Bluetooth Connection	
		Select Components and connect properly

Appendix E. Device Test Plan

Table E1: Test Descriptions

Test ID	Test Description	Expected Results	Actual Results	Pass/Fail
Linear Actuator Functional Test	Testing whether the linear actuator can reliably move up and down at the specified rate for a period of 10 – 15 minutes continuously.	Expected to pass the 10 – 15-minute test as the motor has shown previous reliability.	The actuator was able to move up and down at a consistent pace for 10-15 minutes.	Pass
Linear Actuator Performance Test	Testing the speed of which the linear actuator can reliably move at for an extended period of about 2 – 3 minutes.	Expected to pass but may have some issues develop over time as the lead screw begins to wear.	The actuator is able to fully extend and retract without any issues.	Pass
Linear Actuator Durability Test	Testing the linear actuator at constantly varying speeds and time intervals to see the durability and reliability of extended use.	Expected to pass but may struggle when transitioning from a slow rate to a higher rate. Additional optimizations for slow transition between rates may be necessary.	Since in our project we are going based on travel time and not actual speed. It can move at a consistent speed for a long time.	Pass
Heart Rate Lighting Test	Testing the accuracy of the heart rate detection system under different lighting conditions varying from evening ambient lighting to lighting in a bright room.	The results of this test may vary as lighting seems to have a profound effect on the results of the heart rate and optimization is still being done.	Full daylight settings resulted in the camera being blown out and the heart rate losing accuracy, however all indoor conditions and lower outdoor lighting conditions give an accurate heart rate.	Pass
Heart Rate Movement Test	Testing the accuracy of the heart rate detection system while a subject is moving within the capture frame.	Expected to pass as optimizations have been implemented to counter subject movement and increase accuracy with moving targets.	Having large movements results in the heart rate being slightly skewed because of loss of tracking and motion blur, biasing the heart rate	Pass

			towards previously detected frequencies, however smaller movements are handled well, and an accurate heart rate is detected.	
Sound System Volume	Audio will be played multiple times and will be graphed on a waveform to have the amplitude observed.	The amplitude should not change to ensure that the audio doesn't increase past the max volume in place	The sounds that were coming through the speakers were sufficient and consistent whenever an audio file was summoned.	Pass
Sound System Accuracy	Numerous values that represent the heart rate will be sent over to the mp3 module that will then play a designated audio file from memory.	The designated audio file will play that matches the heart rate given	After doing 100 trials with randomized values for heart rates, it came back at 98 percent accuracy.	Pass
Sound System Timing	An audio file will play once the value for the heart rate is known.	The audio should play almost instantaneously once a heart rate has been captured.	When given a dummy value for a heart rate, the program would take on average about 2 seconds to play to audio that we wanted.	Pass

Appendix F. Market Analysis

Table F1: SWOT Analysis Table

Strengths	Weaknesses
• Interactive • Touchless • Bluetooth/ Wi-Fi Compatible • Modular • Application Integration	• Large • Expensive • Heavy • Loud
Opportunities	Threats
• Art Galleries/ Museums • Office Lounges/ Break Rooms • Waiting Rooms • Public Displays • Home Décor • Furniture Integration	• Harsh Outdoor Conditions • Limited Space • Transportation Costs • Installation Costs • Portability Issues